

Dr. B. C. Roy Engineering College, Durgapur

Department of Electrical Engineering
B. Tech (EE) | 5th Semester EE2 | Autonomy

Subject: Power Electronics Subject Code: EE-504

Teaching-Learning Method: Think-Pair-Share (TPS)

Date: Tuesday, 1st September 2026 Time: 10:00–10:50 AM Section: EE-2

Faculty: Kingsuk Majumdar, Ph.D. (Engg.) Dept. of EE, BCREC

i Topic: DC-DC Converters – Step-Down (Buck) Chopper and Step-Up (Boost) Chopper Analysis **Module:** 4

CO: CO2, CO4 **Bloom's Level:** BL3 (Apply) for parts (a) and (b) | BL4 (Analyse) for part (c)

Reference: M. H. Rashid, *Power Electronics*, PHI, 3rd Ed., Ch. 5; P. S. Bimbhra, *Power Electronics*, Khanna Publishers, 3rd Ed., Ch. 6.

NPTEL: Dr. G. Narayanan, IISc Bangalore, *Power Electronics*, <https://nptel.ac.in/courses/108/108/108108035/>

Instructions to Students:

- This activity has three phases: **Think** (8 minutes), **Pair** (12 minutes), and **Share** (10 minutes). Remaining 10 minutes are for faculty wrap-up and consolidation.
- During the **Think** phase, attempt the problem *individually* without consulting your neighbour. Write your answer on the **individual response sheet** provided separately.
- During the **Pair** phase, discuss your approach with your assigned partner, reconcile any differences, and write the combined solution on the **pair response sheet**.
- During the **Share** phase, selected pairs will present their solution on the board. Active participation is expected from the entire class.
- **Both sheets** (individual and pair) must be submitted before leaving the class. These are mandatory NBA file documents.
- Write your **Name, Roll Number, and Pair Number** clearly on every sheet.
- Carry your scientific calculator and pen.

⚡ TPS Question – DC-DC Chopper Analysis

Problem Statement:

A DC-DC chopper circuit operates from a supply voltage of $V_s = 200$ V. The load consists of a resistance $R = 8 \Omega$ and an inductance $L = 20$ mH. The chopper operates at a switching frequency of $f = 1$ kHz. The duty ratio is $\alpha = 0.6$.

(a) [4 marks] For the **Step-Down (Buck) Chopper**:

- Write the expression for the average output voltage V_o and compute its value.
- Compute the RMS output voltage $V_{o,rms}$.
- Compute the average output current I_o .
- Compute the peak-to-peak ripple current ΔI_L in the inductor, assuming continuous conduction mode (CCM).

State and draw the equivalent circuit clearly, identifying the on-state and off-state paths.

(b) [3 marks] For the **Step-Up (Boost) Chopper** with the same supply $V_s = 200$ V and duty ratio $\alpha = 0.6$:

- Derive the expression for the output voltage V_o in terms of V_s and α .
- Compute the output voltage V_o .
- If the output current (load current) is $I_o = 10$ A, compute the average input current (source current) I_s .
- Comment on the power balance between input and output.

(c) [3 marks] **Comparison and Analysis:**

- Using the results of parts (a) and (b), compare the output voltages of the buck and boost converters for the same $V_s = 200$ V and $\alpha = 0.6$. Express both as a ratio of V_s .
- For the buck chopper in part (a), if the duty ratio is increased to $\alpha = 0.8$ while keeping all other parameters unchanged, determine the new average output voltage and the new peak-to-peak ripple current ΔI_L .
- State **one practical application** each for the step-down chopper and the step-up chopper in electrical systems.

💡 Hint (refer only if stuck):

For a *step-down chopper*: $V_o = \alpha V_s$ (average), $V_{o,rms} = \sqrt{\alpha} V_s$, $\Delta I_L = \frac{V_s \alpha (1 - \alpha)}{f L}$

For a *step-up chopper*: $V_o = \frac{V_s}{1 - \alpha}$

Power balance (lossless): $P_{in} = P_{out}$, i.e. $V_s I_s = V_o I_o$.

Total Marks: 10

Think Phase: 8 min

Pair Phase: 12 min

Share

Phase: 10 min

🔑 **Model Solution – For Faculty Reference Only (Not for Distribution)**

Part (a): Step-Down (Buck) Chopper

Given: $V_s = 200$ V, $R = 8$ Ω , $L = 20$ mH, $f = 1$ kHz, $\alpha = 0.6$.

(i) Average output voltage:

$$V_o = \alpha V_s = 0.6 \times 200 = \mathbf{120} \text{ V}$$

(ii) RMS output voltage:

$$V_{o,rms} = \sqrt{\alpha} V_s = \sqrt{0.6} \times 200 = 0.7746 \times 200 = \mathbf{154.9} \text{ V}$$

(iii) Average output current:

$$I_o = \frac{V_o}{R} = \frac{120}{8} = \mathbf{15} \text{ A}$$

(iv) Peak-to-peak inductor ripple current (CCM):

$$\Delta I_L = \frac{V_s \alpha (1 - \alpha)}{f L} = \frac{200 \times 0.6 \times 0.4}{1000 \times 20 \times 10^{-3}} = \frac{48}{20} = \mathbf{2.4} \text{ A}$$

Part (b): Step-Up (Boost) Chopper

(i) Derivation of output voltage:

During the on-period ($T_{on} = \alpha T$), the switch is closed and the inductor stores energy; the diode is reverse biased and the load is supplied by the capacitor. During the off-period ($T_{off} = (1 - \alpha)T$), the switch opens; the inductor releases energy through the diode to the load. Applying volt-second balance on the inductor:

$$V_s \alpha T = (V_o - V_s)(1 - \alpha) T \Rightarrow V_o = \frac{V_s}{1 - \alpha}$$

(ii) Output voltage:

$$V_o = \frac{200}{1 - 0.6} = \frac{200}{0.4} = \mathbf{500} \text{ V}$$

(iii) Average input (source) current:

From power balance (lossless converter): $V_s I_s = V_o I_o$

$$I_s = \frac{V_o I_o}{V_s} = \frac{500 \times 10}{200} = \mathbf{25} \text{ A}$$

(iv) Power balance:

$$P_{in} = V_s \times I_s = 200 \times 25 = 5000 \text{ W} \quad P_{out} = V_o \times I_o = 500 \times 10 = 5000 \text{ W}$$

Input power equals output power, confirming the lossless assumption.

Part (c): Comparison and Analysis

(i) Output voltage comparison:

Converter	Output Voltage	Ratio V_o/V_s
Buck (step-down)	120 V	$\alpha = 0.6$
Boost (step-up)	500 V	$1/(1 - \alpha) = 2.5$

The buck output is a fraction of V_s ; the boost output is a multiple of V_s . For $\alpha = 0.6$, the boost converter amplifies the input voltage by a factor of 2.5.

(ii) Buck chopper with $\alpha = 0.8$:

$$V_o' = 0.8 \times 200 = \mathbf{160 \text{ V}}$$

$$\Delta I_L' = \frac{200 \times 0.8 \times 0.2}{1000 \times 20 \times 10^{-3}} = \frac{32}{20} = \mathbf{1.6 \text{ A}}$$

Note: Increasing α increases the average output voltage but *reduces* the ripple current. Maximum ripple occurs at $\alpha = 0.5$.

(iii) Practical applications:

- **Buck chopper:** DC motor speed control (traction drives, EV drives), battery charging regulation.
- **Boost chopper:** Power factor correction (PFC) front ends, UPS systems, renewable energy (PV panel output boosting for grid interface).

✔ **Answer Summary:**

Part (a): $V_o = 120 \text{ V}$, $V_{o,rms} = 154.9 \text{ V}$, $I_o = 15 \text{ A}$, $\Delta I_L = 2.4 \text{ A}$

Part (b): $V_o = 500 \text{ V}$, $I_s = 25 \text{ A}$, $P_{in} = P_{out} = 5000 \text{ W}$

Part (c): $V_o'(\alpha = 0.8) = 160 \text{ V}$, $\Delta I_L'(\alpha = 0.8) = 1.6 \text{ A}$

All numerical answers verified. Full solutions above.

Dr. B. C. Roy Engineering College, Durgapur

Department of Electrical Engineering

CO-PO Mapping of Think-Pair-Share Activity

Subject: Power Electronics (EE-504) Section: EE-2 Date: 1st September 2026

1 Course Outcomes Addressed

CO2: Analyze and design different types of power electronic converters and their control circuits.

CO4: Design and implement different control strategies for DC-DC converters and inverters.

Bloom's Level: BL3 (Apply) for parts (a) and (b) | BL4 (Analyse) for part (c)

Topic: DC-DC Converters – Step-Down (Buck) Chopper and Step-Up (Boost) Chopper: analysis of output voltage, ripple current, power balance, and comparative performance.

References: M. H. Rashid, *Power Electronics*, PHI, 3rd Ed., Ch. 5; P. S. Bimbhra, *Power Electronics*, Khanna Publishers, 3rd Ed., Ch. 6; Daniel W. Hart, *Power Electronics*, McGraw-Hill, Ch. 7.

NPTEL: Dr. G. Narayanan, IISc Bangalore – <https://nptel.ac.in/courses/108/108/108108035/>

2 CO-PO Mapping Table

CO	Course Outcome	Programme Outcomes (PO)											PSO		
		1	2	3	4	5	6	7	8	9	10	11	1	2	3
CO2	Analyze and design power electronic converters and control circuits	3	3	3	2	3	1	1	1	2	2	2	3	3	1
CO4	Design and implement control strategies for DC-DC converters and inverters	3	3	3	2	3	2	1	2	2	2	2	3	3	1

Table 1: CO-PO/PSO Correlation for TPS Activity (3: High, 2: Medium, 1: Low, -: No correlation), as per EE-504 lesson plan CO-PO mapping

3 Part-wise CO-PO Justification

Part	Skill / Concept Tested	CO	PO	BL
(a-i)	Apply the duty-ratio expression to compute average output voltage of a buck chopper	CO2, CO4	PO1, PO2	BL3
(a-ii)	Apply the RMS voltage expression; distinguish between average and RMS output	CO2	PO1, PO2	BL3
(a-iii)	Compute average output current using Ohm's law on the average output voltage	CO2, CO4	PO1, PO2	BL3
(a-iv)	Apply the CCM ripple current formula; understand the role of L , f and duty ratio on ripple	CO2, CO4	PO1, PO2, PO3	BL3
(b-i)	Derive the boost converter voltage gain using volt-second balance on the inductor	CO2, CO4	PO1, PO2	BL3
(b-ii,iii,iv)	Compute output voltage and input current; verify power balance; understand energy flow in lossless converter	CO2, CO4	PO1, PO2, PO5	BL3
(c-i,ii)	Compare buck and boost performance metrics; evaluate the effect of duty ratio change on ripple and voltage (maximum ripple at $\alpha = 0.5$)	CO2, CO4	PO1, PO2, PO3	BL4
(c-iii)	Identify and justify real-world applications of buck and boost converters in electrical systems	CO4, CO6	PO2, PO6	BL4

Table 2: Part-wise skill, CO, PO and Bloom's Level mapping

4 PO Description Reference

PO	Description (AICTE/NBA)
PO1	Engineering Knowledge: Apply knowledge of mathematics, science, and engineering fundamentals to solve power electronics problems
PO2	Problem Analysis: Identify, formulate, and analyse complex engineering problems using converter circuit equations
PO3	Design / Development of Solutions: Evaluate design choices (duty ratio, inductance, switching frequency) against performance requirements
PO5	Modern Tool Usage: Use analytical methods and simulation tools (eSim / LTspice) for power converter analysis
PO6	The Engineer and Society: Apply power electronics knowledge to real-world industrial and energy applications

Table 3: Relevant PO descriptions for CO2 and CO4 mapping

Dr. B. C. Roy Engineering College, Durgapur

Department of Electrical Engineering

Faculty Evaluation Report

Teaching-Learning Method: Think-Pair-Share (TPS)

Subject: Power Electronics (EE-504) Section: EE-2

Date: Tuesday, 1st September 2026 Time: 10:00–10:50 AM

Faculty: Kingsuk Majumdar, Ph.D. (Engg.) Dept. of EE, BCREC

5 Activity Details

Activity	Think-Pair-Share (TPS)
Subject / Code	Power Electronics / EE-504
Section	EE-2 (EE2X + EE2Y), B. Tech (EE), 5th Semester
Batch	2024–28
Session	2026–27 (ODD Semester)
Date and Time	Tuesday, 1st September 2026, 10:00–10:50 AM
Venue	_____
Total Students Present	_____ out of 47
Total Pairs Formed	_____ (odd student, if any, forms a group of 3)
Topic	DC-DC Converters – Step-Down (Buck) Chopper and Step-Up (Boost) Chopper
Module	Module 4 – DC-DC Converters (Chopper)
CO Addressed	CO2 (Analyze and design converters), CO4 (Design DC-DC converter control strategies)
Bloom's Level	BL3 (Apply) for parts (a) and (b) BL4 (Analyze) for part (c)

Table 4: TPS Activity Details – EE-504

6 ? Question Asked

A DC-DC chopper operates from $V_s = 200$ V, $R = 8\ \Omega$, $L = 20$ mH, $f = 1$ kHz, duty ratio $\alpha = 0.6$.

(a) For the **step-down (buck) chopper**, compute: V_o , $V_{o,rms}$, I_o , and peak-to-peak ripple current ΔI_L (CCM assumed). Draw the equivalent circuit. [4 marks]

(b) For the **step-up (boost) chopper** with same V_s and α : derive V_o , compute V_o , find I_s if $I_o = 10$ A, and verify power balance. [3 marks]

(c) Compare V_o of both converters as a ratio of V_s ; compute new V_o and ΔI_L for buck with $\alpha = 0.8$; state one practical application each. [3 marks]

Reference: M. H. Rashid, *Power Electronics, PHI, 3rd Ed.*; P. S. Bimbhra, *Power Electronics, Khanna Publishers*.

7 🕒 Phase-wise Time Observation

Phase	Duration	Time Slot	Faculty Observation
Think	8 minutes	10:00–10:08 AM	_____
Pair	12 minutes	10:08–10:20 AM	_____
Share	10 minutes	10:20–10:30 AM	_____
Wrap-up	20 minutes	10:30–10:50 AM	Faculty consolidation: buck/boost expressions, ripple analysis, maximum-ripple condition ($\alpha = 0.5$), and applications.

Table 5: Phase-wise time and observation log

8 ⚠️ Common Mistakes / Misconceptions to Watch

Fill after conducting the activity. Typical mistakes to watch for:

- Average vs. RMS output voltage:** Students use $V_o = \alpha V_s$ as both average and RMS. The RMS is $V_{o,rms} = \sqrt{\alpha} V_s \neq \alpha V_s$. For $\alpha = 0.6$: average = 120 V, RMS = 154.9 V. This is a standard GATE error. _____ (observed: Yes / No)
- Ripple current formula sign / factor error:** Students write $\Delta I_L = V_o T_{off}/L$ or mix up T_{on} and T_{off} in the formula. The correct CCM expression is $\Delta I_L = V_s \alpha (1 - \alpha)/(fL)$. _____ (observed: Yes / No)
- Boost converter derivation – energy direction:** Students confuse which period charges

and which period discharges the inductor. The volt-second balance approach must be clearly demonstrated: $V_s T_{on} = (V_o - V_s) T_{off}$. _____ (observed: Yes / No)

4. **Power balance in boost converter:** Students compute I_s by dividing I_o by the voltage gain, getting $I_s = I_o(1 - \alpha)$ – the correct relation. Errors occur when students invert the ratio. _____ (observed: Yes / No)

5. **Maximum ripple condition:** Students incorrectly state that maximum ripple occurs at $\alpha = 1$. From $\Delta I_L = V_s \alpha (1 - \alpha) / (fL)$, differentiating w.r.t. α and setting to zero gives $\alpha = 0.5$ for maximum ripple. _____ (observed: Yes / No)

6. **Additional mistakes observed during class:**

9 Performance Summary

Criterion	Excellent	Satisfactory	Needs Improvement
Individual (Think) response quality			
Pair discussion depth and correction			
Correctness of part (a) – buck analysis			
Correctness of part (b) – boost derivation			
Correctness of part (c) – comparison and analysis			
Active participation in Share phase			
Understanding of average vs. RMS distinction			

Table 6: Class performance assessment summary (tick appropriate column)

10 Pairs Selected for Share Phase (Board Presentation)

Sl.	Student 1 (Name / Roll)	Student 2 (Name / Roll)	Part Presented
1			Part (a)
2			Part (b)
3			Part (c)

Table 7: Pairs selected for board presentation during Share phase

11 ★ Key Learning Points Emphasised During Wrap-up

1. The duty ratio α is the single most powerful design parameter in a DC chopper. For a buck converter, $V_o = \alpha V_s$ gives continuous, linear control of output voltage from zero to V_s . For a boost converter, $V_o = V_s/(1 - \alpha)$ gives theoretically unbounded gain as $\alpha \rightarrow 1$, but in practice losses and component ratings limit the maximum duty ratio.
2. The distinction between average output voltage (αV_s) and RMS output voltage ($\sqrt{\alpha} V_s$) in a buck chopper is essential for correct power and current calculations. The average voltage drives the DC component of load current; the RMS voltage determines the total power dissipated in a resistive load.
3. The peak-to-peak ripple current formula $\Delta I_L = V_s \alpha (1 - \alpha)/(fL)$ shows that ripple is maximised at $\alpha = 0.5$ and can be reduced by increasing f (switching frequency) or L (inductance). This is the fundamental design trade-off in chopper circuits.
4. Volt-second balance is the universal tool for deriving the steady-state voltage gain of any DC-DC converter operating in CCM. It states that the net volt-seconds applied to an inductor over one complete switching period must be zero in steady state.
5. Additional points noted during class:

12 Faculty Remarks and Overall Assessment

Overall observation of individual (Think) phase:

Overall observation of Pair discussion quality:

Overall observation of Share phase presentations:

Effectiveness of TPS methodology for this topic:

Action to be taken in next class (remedial / follow-up):

Kingsuk Majumdar, Ph.D. (Engg.)

Assistant Professor, Dept. of EE
Dr. B. C. Roy Engineering College, Durgapur

Date: _____

Signature: _____

13 NBA 2 TLM – Think-Pair-Share Activity (EE-504)

13.1 Activity Overview

TLM Method:	Think-Pair-Share (TPS)
Subject / Code:	Power Electronics / EE-504
Section:	EE-2 (EE2X + EE2Y), B. Tech (EE), 5th Semester, Batch 2024–28
Date and Time:	Tuesday, 1st September 2026, 10:00–10:50 AM
Faculty:	Kingsuk Majumdar, Ph.D. (Engg.), Dept. of EE, BCREC
Topic:	DC-DC Converters – Step-Down (Buck) and Step-Up (Boost) Chopper Analysis
Module:	Module 4 – DC-DC Converters (Chopper)
CO Addressed:	CO2 – Analyze and design power electronic converters and their control circuits; CO4 – Design and implement control strategies for DC-DC converters
Bloom's Level:	BL3 (Apply) for parts (a) and (b), BL4 (Analyse) for part (c)
Total Students:	47 (EE2X: 24 students, EE2Y: 23 students)

13.2 Pair List – EE-504 TPS Activity

Pair	Roll No.	Name	Roll No.	Name
1	120016244052	Soham Bhat-tacharjee	120016244053	Soham Ganguly
2	120016244054	Souronil Kundu	120016244055	Souvik Ganguli
3	120016244056	Subha Sarkar	120016244057	Subham Dutta
4	120016244058	Subham Kundu	120016244059	Sudip Maity
5	120016244060	Suman Pramanik	120016244061	Sundram Kumar
6	120016244062	Surjendu Mukherjee	120016244064	Tanbir Haque Mondal
7	120016244065	Tirtha Mondal	120016244066	Tushar Roy
8	120016244067	Siddhartha Dey	120016253001	Abhirup Adhikary
9	120016253002	Abhishek Ghoshal	120016253003	Adarsha Maji
10	120016253004	Anirudra Das	120016253005	Anish Rudra
11	120016253006	Ankesh Tanti	120016253007	Ayush Roy
12	120016253008	Bibek Roy	120016253009	Bikash Shit
13	120016253010	Debdeep Mondal	120016253011	Jeet Sain
14	120016253012	Jit Ghosh	120016253013	Koushik Pal
15	120016253014	Krishn Kumar Sharma	120016253015	Kumar Aditya
16	120016253016	Mangalam Thakur	120016253017	Mirza Amir Baig
17	120016253018	Nitish Kumar	120016253019	Nityapriyo Roy
18	120016253020	Probal Chandra Biswas	120016253021	Raima Podder
19	120016253022	Rajib Pandit	120016253023	Ramen Ruidas
20	120016253024	Sachin Turi	120016253025	Sandip Biswas

Pair	Roll No.	Name	Roll No.	Name
21	120016253026	Sarmistha Pra- manik	120016253028	Sayan Debnath
22	120016253029	Sayan Gorai	120016253030	Subhadip Mondal
23	120016253031	Sudip Pramanik	120016253032	Sujoy Ghosh
<i>Pair 24 (Group of 3 – odd student):</i>				
24	120016253033	Trideb Ghosh	(joins Pair 23 to form a group of 3)	

i Note: Total 47 students. 23 pairs (Pairs 1–23) of 2 students each + 1 group of 3 (Pair 24: Trideb Ghosh joins Pair 23). Roll 120016244063 is absent from the roll register.

13.3 Rationale for Topic Selection

Module 4 (DC-DC Converters – Chopper) is the second module delivered in the EE-504 course, completed during Weeks 2 and 3 (ending 17-August-2026). By 1st September 2026 (Week 5), students have had two full weeks to consolidate their understanding of chopper circuits from lectures. The buck and boost chopper analysis was selected for the TPS activity for the following reasons. The three-part structure of the question maps naturally onto the three TPS phases, with progressively increasing cognitive demand: part (a) tests direct formula application (BL3), part (b) requires derivation and power balance reasoning (BL3), and part (c) demands comparison and evaluation (BL4). Chopper numericals involve a self-contained set of equations that students can attempt individually in the Think phase and meaningfully resolve disagreements in the Pair phase (particularly the RMS vs. average voltage distinction, and the volt-second balance derivation). The Share phase board presentations naturally distribute the three parts across pairs. The activity also serves as a focused revision for Class Test-1 (09-September-2026), which covers Modules 1, 4, and 2.

13.4 Phase-wise Plan

Phase	Duration	Time	Description
Think	8 min	10:00–10:08 AM	Each student attempts all three parts individually on the response sheet. No peer interaction is permitted. Faculty circulates silently to observe individual engagement and note which parts students attempt first (typically (a-i) and (a-ii) are attempted immediately; the boost derivation in (b-i) is usually the hesitation point).
Pair	12 min	10:08–10:20 AM	Students discuss with their assigned partner (from the pair list in Section 3.2 above). They reconcile differences, particularly on: (i) whether the ripple formula uses V_s or V_o ; (ii) the direction of the volt-second balance for the boost converter; and (iii) the maximum ripple condition. The combined solution is written on the pair response sheet. Faculty observes and notes systematic errors across pairs.
Share	10 min	10:20–10:30 AM	Three pairs are called to present one part each on the board. The class critiques each solution. Faculty moderates, corrects errors in the RMS formula and volt-second balance immediately, and confirms the correct answers.
Wrap-up	20 min	10:30–10:50 AM	Faculty consolidates: (i) buck and boost expressions with derivation; (ii) maximum ripple condition at $\alpha = 0.5$; (iii) power balance in lossless converters; (iv) practical applications in EV drives, PFC, and UPS. Connects to Class Test-1 (09-Sep-2026) scope.

Table 9: Phase-wise execution plan for TPS activity – EE-504

13.5 CO-PO Mapping Summary

CO	PO1	PO2	PO3	PO5	PSO1	PSO2	PSO3	Bloom's Level
CO2	3	3	3	3	3	3	1	BL3, BL4
CO4	3	3	3	3	3	3	1	BL3, BL4

Table 10: CO-PO-PSO correlation for TPS activity (3: High, 2: Medium, 1: Low)

13.6 Supporting Documents for NBA File Submission

Sl.	Document	Format
1	Pair list with Name, Roll No., Pair No. (Table in Section 3.2 of this document)	PDF
2	TPS Question Sheet (Document 1, pages 1–2 of this file)	PDF
3	Individual (Think) response sheets – minimum 10 samples	Single PDF
4	Pair (combined) response sheets – minimum 10 samples	Single PDF
5	Score / marks sheet (filled during evaluation)	PDF
6	Photographs: students working individually and in pairs (GEO tag + non-GEO tag)	Single PDF
7	Faculty Evaluation Report (Document 3, this file)	PDF
8	CO-PO mapping of activity (Document 2, this file)	PDF

Table 11: Documents to be submitted for TPS NBA 2 file – EE-504

Document Prepared by: Kingsuk Majumdar, Ph.D. (Engg.)
 Department of Electrical Engineering
 Dr. B. C. Roy Engineering College, Durgapur

Date: 30-August-2026 **Version:** V2

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